

Fundamentals of Information Security

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Basic Setup:

- Communication over an insecure channel
- Types of insecure channel
 - Internet (unprotected network of computers)
 - Wifi (not password protected)
 - Air Waves (GSM connection) etc.

Meet Alice and Bob



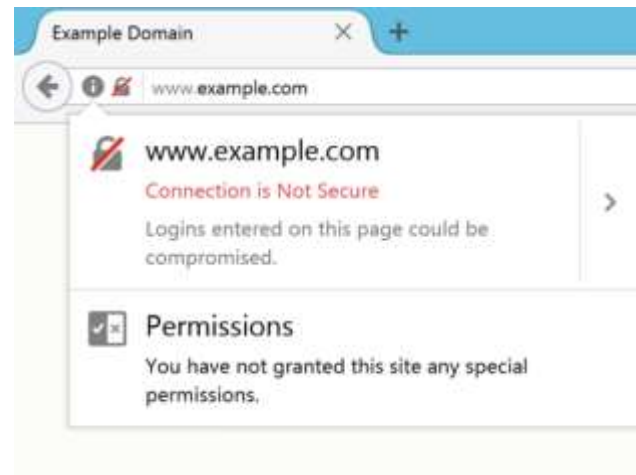
ALICE

Message
Username: sweetalice
Password: alice123
TRANSFER Rs. 100 TO
ACCOUNT NO. XYZ

Insecure network channel



BOB at Bank



**NOT
ENCRYPTED**

Meet Charlie – the eavesdropper



ALICE

Message
Username: sweetalice
Password: alice123
TRANSFER Rs. 100 TO
ACCOUNT NO. XYZ



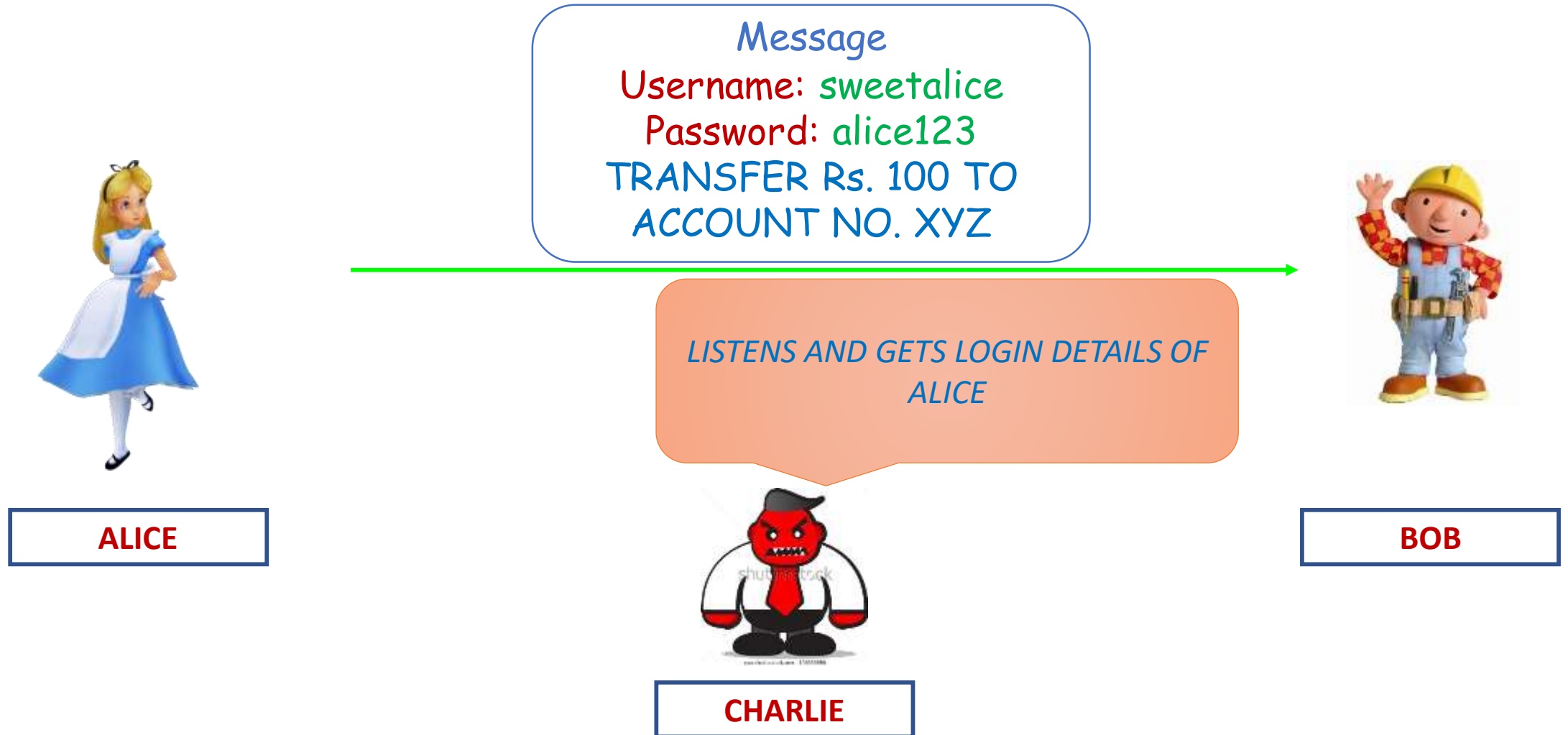
BOB

Listen

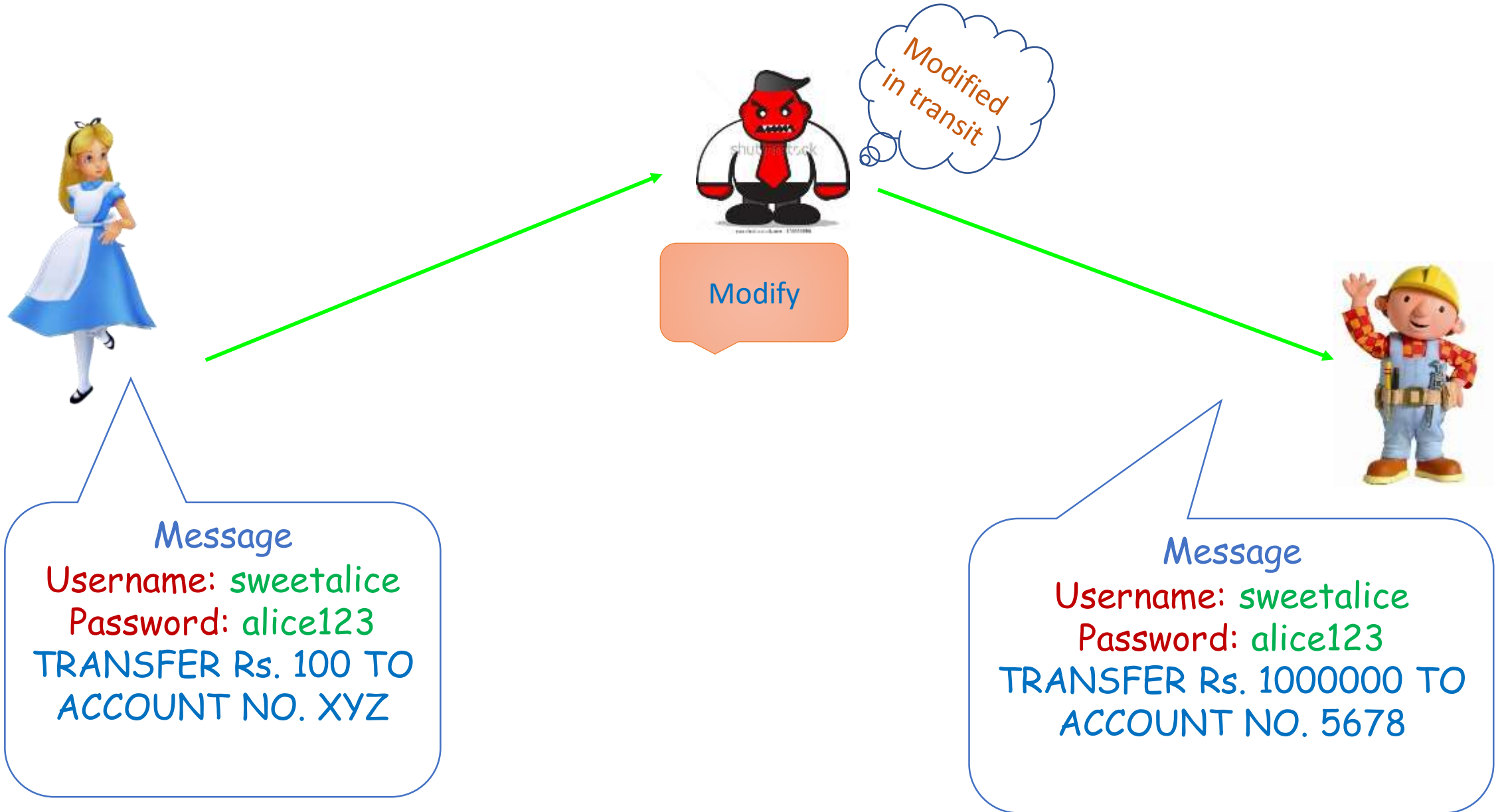


CHARLIE

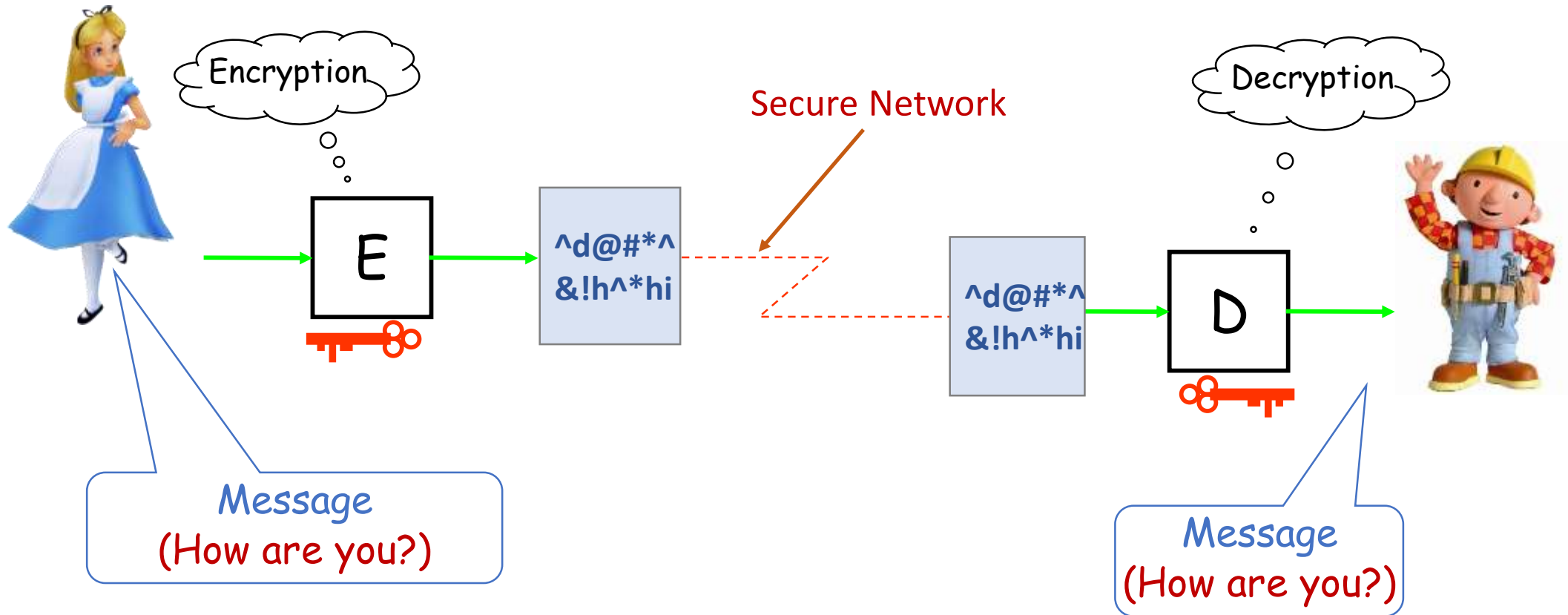
Username, Password sent in clear!!!



Username, Password sent in clear!!!



Confidentiality can be achieved using encryption/decryption



E: Encryption- Charlie cannot see what is being sent over the channel
D: Decryption- Bob can successfully decrypt the message

Basic Definitions:

- Plaintext (P) – The original message
- Ciphertext (C) – The scrambled message
- $E()$ – Encryption Function
- $D()$ – Decryption Function

Basic Definitions:

Q. To ensure safety of our data, should the enc./dec. algorithm be known to all or kept secret ??

Why secrecy of algorithms is not a good idea ?

- Keeping an algorithm secret is difficult because competitors can steal it, employees can leak it, and attackers can analyze the encryption software to discover it.
- That's why modern cryptography generally says: **make the algorithm public and keep the key secret.**

But ...

- We can't make everything public
 - Eve can easily decrypt then
- Incorporate a second parameter
 - The "KEY"
 - Short secret data shared by both communicating parties



Kerckhoffs's Principle [1883]:

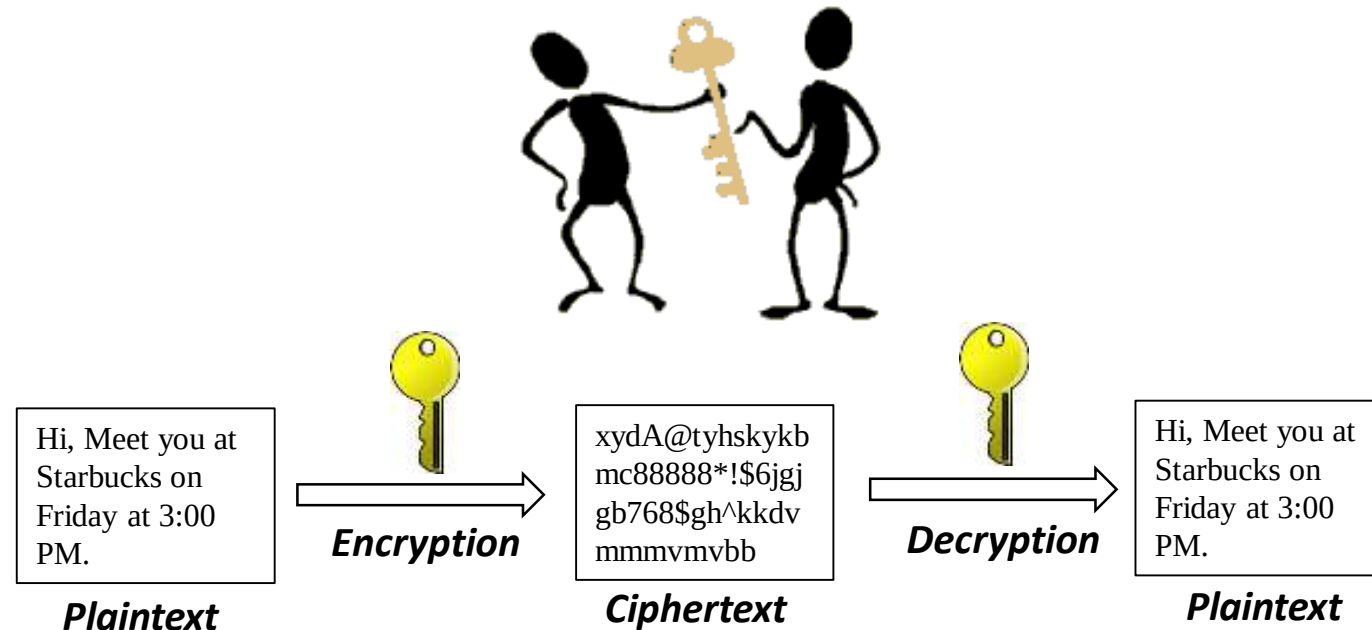
A cryptosystem should be secure even if the attacker knows all the details of the system, with the exception of the secret key.

Depending upon the key, cryptosystems can be divided into:

- Symmetric Cryptosystems
- Asymmetric Cryptosystems

Symmetric Encryption

- Both the parties use the **same key** to encrypt and decrypt

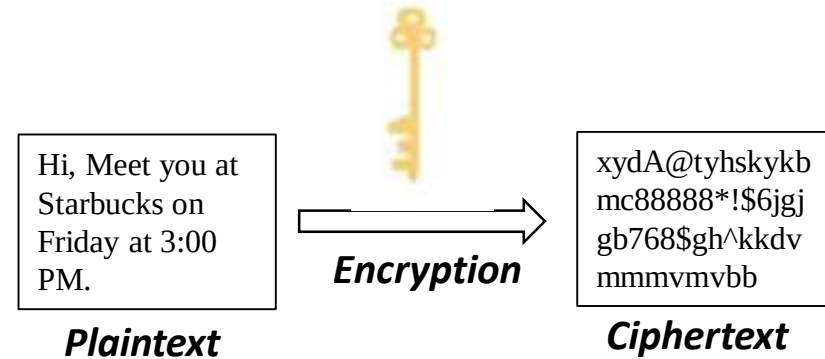


Asymmetric Encryption

- Communicating parties use **two keys** – **public key (known to all)** and **private key (known only to owner)** to encrypt and decrypt



1. Bob gives Alice his public key



2. Alice uses the public key to encrypt her message

Asymmetric Encryption

- Communicating parties use **two keys** – **public key (known to all)** and **private key (known only to owner)** to encrypt and decrypt



3. Bob receives Alice's encrypted message

4. Bob uses his private key to decrypt her message

Asymmetric Encryption

- Can the reverse be used, i.e., private key is used for encryption and public key is used for decryption ?
 - Why or why not ?

How do we keep this key secret ?

- Either Alice and Bob meet in private
 - Exchange the keys used for later communications
- Use special cryptographic key exchange protocol
 - Diffie Hellman Key Exchange